

Iterative Cameron–Martin Reflections Couple Banach Brownian Motions at Time Infinity

Candidate solution packet for arXiv:1705.08300

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Status. Candidate full solution, likely valid pending human review. The main review point is the standard predictable-reflection assertion used to preserve the Brownian marginal during the stage-by-stage construction.

1 Source and open problem

The source paper is Candellero–Kendall, *Coupling of Brownian motions in Banach spaces*, arXiv:1705.08300 [1]. Its Question 18 asks whether, given an abstract Wiener space

$$\mathcal{W}^* \hookrightarrow \mathcal{H} \hookrightarrow \mathcal{W},$$

two $\mathcal{W}$ -valued Brownian motions started from arbitrary starting points in $\mathcal{W}$ can always be coupled at time infinity. In the notation of the paper, this means constructing a coupling such that

$$\left\| \tilde{B}(t) - B(t) \right\|_{\mathcal{W}} \longrightarrow 0 \quad \text{almost surely as } t \rightarrow \infty.$$

The crop below shows the source question.

This paper has shown that Brownian couplings at time ∞ are always possible if $\mathcal{W}$ supports a Schauder basis which is also $\mathcal{H}_\mu$ -orthogonal (Theorem 13), and in particular that they are always possible in the important special case when $\mathcal{W}$ is a Hilbert space (Lemma 15). It is therefore natural to ask

Question 18. Given an abstract Wiener space $\mathcal{W}^* \hookrightarrow \mathcal{H}_\mu \hookrightarrow \mathcal{W}$, is it always possible to produce a coupling at time ∞ for two $\mathcal{W}$ -valued Brownian motions started from arbitrary starting points in $\mathcal{W}$? (Or if not, then what Banach-space geometry property for $\mathcal{W}^* \hookrightarrow \mathcal{H}_\mu \hookrightarrow \mathcal{W}$ corresponds to the probabilistic property that Brownian coupling at time ∞ is always possible?)

Certainly there are abstract Wiener spaces possessing the property of Brownian coupling at time ∞ which do not possess a $\mathcal{H}_\mu$ -orthogonal Schauder basis. Recall that $\mathcal{W}$ has the finite dimensional decomposition property (FDD) if it supports finite-dimensional subspaces $\mathcal{W}^{(n)}$ such that $x = \sum_{n=1}^{\infty} x_n$ (conditionally convergent) for unique $x_n \in \mathcal{W}^{(n)}$ to be compatible with the Cameron–Martin geometry

The source paper proves this under additional Schauder-basis, FDD, or orthogonal M-basis hypotheses. The construction below uses only the density of the Cameron–Martin space $\mathcal{H}$ in $\mathcal{W}$, which is part of the abstract Wiener space structure.

2 Local source ingredient

The proof iterates the paper’s Cameron–Martin reflection coupling. For $h \in \mathcal{H} \setminus \{0\}$, the reflection on $\mathcal{H}$ is

$$R_h(y) = y - 2 \frac{\langle h, y \rangle_{\mathcal{H}}}{\|h\|_{\mathcal{H}}^2} h.$$

Candellero–Kendall explain that this reflection produces a reflected $\mathcal{W}$ -valued Brownian motion without requiring R_h to extend to a bounded operator on $\mathcal{W}$; one decomposes the abstract Wiener space along $\mathbb{R}h \oplus h^\perp$. The relevant definition appears here.

of $\mathbf{B}(t)$ and $\tilde{\mathbf{B}}(t)$ are mutually singular, and therefore $\mathbb{P}[\tilde{\mathbf{B}} \text{ meets } \mathbf{B} \text{ by time } t] = 0$, for all t and all possible couplings. On the other hand, if x does lie in $\mathcal{H}_\mu$ then there is a *maximal* successful coupling: maximal in the sense that (3.1) becomes an equality for all t .

3.1 Cameron–Martin Reflections in Banach spaces

Given $x \in \mathcal{H}_\mu$, construct the $\mathcal{H}_\mu$ -projection $h \mapsto \langle x, h \rangle_{\mathcal{H}_\mu} x / \|x\|_{\mathcal{H}_\mu}^2$. Define the resulting *Cameron–Martin reflection* $R_x(y) : \mathcal{H}_\mu \rightarrow \mathcal{H}_\mu$ by

$$R_x(y) = y - 2 \frac{\langle x, y \rangle_{\mathcal{H}_\mu}}{\|x\|_{\mathcal{H}_\mu}^2} x \quad \text{for } y \in \mathcal{H}_\mu. \quad (3.2)$$

Note that $R_x(x) = -x$. By Theorem 6, R_x produces a *reflected* Brownian motion $\tilde{\mathbf{B}} = x + R_x(\mathbf{B})$: we

3 Proof intuition

The obstruction to finite-time coupling is that the initial displacement may not lie in $\mathcal{H}$. But $\mathcal{H}$ is dense in $\mathcal{W}$. We therefore write the displacement $x \in \mathcal{W}$ as a convergent telescoping sum of Cameron–Martin corrections $h_n \in \mathcal{H}$, with

$$x = h_1 + h_2 + \cdots \quad \text{in } \mathcal{W}$$

and with $\sum_n \|h_n\|_{\mathcal{W}} < \infty$. At stage n , the two Brownian motions differ by a deterministic residual $r_{n-1} = h_n + r_n$. A reflection coupling in the single Cameron–Martin direction h_n removes h_n in finite time, leaving the smaller deterministic residual r_n . Then the motions are run synchronously for one unit of time, and the next stage begins.

The reflected stage can overshoot in the negative h_n -direction before it hits the coupling hyperplane. Its maximum scalar stretch has a heavy one-dimensional gambler’s-ruin tail, but when multiplied by the summable $\mathcal{W}$ -norms $\|h_n\|_{\mathcal{W}}$, Borel–Cantelli forces the stage suprema to tend to zero. Thus the Banach-space distance between the two paths tends to zero along all times, not merely at stage endpoints.

4 The theorem

Theorem 1. *Let $(\mathcal{W}, \mathcal{H}, \mu)$ be an abstract Wiener space, with $\mathcal{H}$ densely and continuously embedded in the separable Banach space $\mathcal{W}$. Let B be a $\mathcal{W}$ -valued Brownian motion whose time-one law is μ . For any two starting points $a, b \in \mathcal{W}$, there is a coupling of $\mathcal{W}$ -valued Brownian motions $B^a, \tilde{B}^b$, started respectively at a and b , such that*

$$\mathbb{P} \left[\left\| \tilde{B}^b(t) - B^a(t) \right\|_{\mathcal{W}} \rightarrow 0 \text{ as } t \rightarrow \infty \right] = 1.$$

Consequently Question 18 of [1] has an affirmative answer.

Proof. By translation invariance it suffices to start one Brownian motion at 0 and the other at $x = b - a \in \mathcal{W}$. If $x = 0$, take the synchronous coupling. Assume $x \neq 0$.

Step 1: a summable Cameron–Martin approximation

Choose a positive summable sequence $(\varepsilon_n)_{n \geq 1}$ tending to zero, for instance $\varepsilon_n = 2^{-n}$. Set $r_0 = x$. Since $\mathcal{H}$ is dense in $\mathcal{W}$, choose inductively $h_n \in \mathcal{H}$ such that

$$r_n := r_{n-1} - h_n, \quad \|r_n\|_{\mathcal{W}} \leq \varepsilon_n.$$

Then $r_n \rightarrow 0$ in $\mathcal{W}$, and

$$r_{n-1} = h_n + r_n.$$

Moreover, for $n \geq 2$,

$$\|h_n\|_{\mathcal{W}} \leq \|r_{n-1}\|_{\mathcal{W}} + \|r_n\|_{\mathcal{W}} \leq \varepsilon_{n-1} + \varepsilon_n.$$

Thus

$$\sum_{n=1}^{\infty} \|h_n\|_{\mathcal{W}} < \infty.$$

Step 2: one reflected reduction

We record the elementary one-step coupling estimate. Fix deterministic $r \in \mathcal{W}$ and $h \in \mathcal{H} \setminus \{0\}$. Suppose at some stopping time the two processes differ by $r + h$. From that time onward, use the Cameron–Martin reflection coupling in the direction h until the one-dimensional component hits the coupling hyperplane, and then run the processes synchronously. During the reflected part of this stage the difference has the form

$$r + \alpha(t)h,$$

where $\alpha(0) = 1$, $\alpha(t) > 0$ before the hitting time, and $\alpha = 0$ after it. The hitting time is finite almost surely because it is the hitting time of a real Brownian motion.

Let

$$A_h = \sup\{\alpha(t) : \text{during this reflected stage}\}.$$

Then for every $M \geq 1$,

$$\mathbb{P}[A_h \geq M] \leq \frac{1}{M}. \tag{1}$$

Indeed, write β for the one-dimensional Brownian component in the direction $h/\|h\|_{\mathcal{H}}$, begun at 0, and stop it on first hitting $\|h\|_{\mathcal{H}}/2$. Before this hit,

$$\alpha(t) = 1 - \frac{2\beta(t)}{\|h\|_{\mathcal{H}}}.$$

Thus $\{A_h \geq M\}$ is the event that β hits $-(M-1)\|h\|_{\mathcal{H}}/2$ before it hits $+\|h\|_{\mathcal{H}}/2$. The usual gambler's-ruin formula for one-dimensional Brownian motion gives probability

$$\frac{\|h\|_{\mathcal{H}}/2}{(M-1)\|h\|_{\mathcal{H}}/2 + \|h\|_{\mathcal{H}}/2} = \frac{1}{M}.$$

Step 3: concatenate the stages

Construct the pair stage by stage. At the beginning of stage n , arrange that the deterministic difference is $r_{n-1} = h_n + r_n$. If $h_n = 0$, skip the reflected part. Otherwise, apply the one-step reflected reduction to remove h_n , leaving difference r_n . Then run the two paths synchronously for one deterministic unit of time before beginning stage $n+1$.

Let σ_n be the end time of stage n , including the one-unit synchronous wait. Since $\sigma_n \geq n$, only finitely many stages occur before any fixed deterministic time, and $\sigma_n \rightarrow \infty$.

Each finite initial segment is a finite concatenation, at stopping times, of ordinary Cameron–Martin reflection couplings and synchronous couplings. By the strong Markov property and the usual invariance of Brownian motion under predictable orthogonal Cameron–Martin reflections, both marginals are $\mathcal{W}$ -valued Brownian motions. Equivalently, for every finite family $\ell_1, \dots, \ell_m \in \mathcal{W}^*$, the corresponding transformed coordinate process is a continuous local martingale with covariance matrix

$$t \left(\text{Cov}_\mu(\ell_i, \ell_j) \right)_{i,j=1}^m,$$

because the integrand is orthogonal on $\mathcal{H}$ at every time; Levy’s characterization gives the required finite-dimensional Brownian projections. The path itself is $\mathcal{W}$ -continuous, since on each finite time interval it is obtained from the original $\mathcal{W}$ -Brownian path by finitely many standard reflections of the form used in [1].

Step 4: convergence at time infinity

Let A_n be the scalar stage supremum from Step 2 for the n -th reflection, with $A_n = 0$ if $h_n = 0$. During the reflected part of stage n ,

$$\left\| \tilde{B}(t) - B(t) \right\|_{\mathcal{W}} \leq \|r_n\|_{\mathcal{W}} + A_n \|h_n\|_{\mathcal{W}},$$

and during the following synchronous wait the distance is exactly $\|r_n\|_{\mathcal{W}}$.

Fix $\eta > 0$. For all sufficiently large n , $\|h_n\|_{\mathcal{W}} < \eta$. Using (1),

$$\mathbb{P}[A_n \|h_n\|_{\mathcal{W}} > \eta] = \mathbb{P}\left[A_n > \frac{\eta}{\|h_n\|_{\mathcal{W}}}\right] \leq \frac{\|h_n\|_{\mathcal{W}}}{\eta}.$$

The right-hand side is summable in n . By Borel–Cantelli,

$$A_n \|h_n\|_{\mathcal{W}} \longrightarrow 0 \quad \text{almost surely.}$$

Since also $\|r_n\|_{\mathcal{W}} \leq \varepsilon_n \rightarrow 0$, the supremum of $\left\| \tilde{B}(t) - B(t) \right\|_{\mathcal{W}}$ over all times in stage n tends to zero almost surely. Because the stage end times tend to infinity, this is exactly

$$\left\| \tilde{B}(t) - B(t) \right\|_{\mathcal{W}} \rightarrow 0 \quad \text{almost surely as } t \rightarrow \infty.$$

This proves the theorem. □

5 Verification and limitations

The proof does not use a Schauder basis, FDD, or M-basis. The only structural input beyond the standard Banach-valued Brownian-motion framework is the dense continuous embedding $\mathcal{H} \hookrightarrow \mathcal{W}$, which is part of the abstract Wiener space setup and is stated in the source paper.

The important technical point for human review is the marginal-preservation step in the infinite stage construction. Since a deterministic one-unit synchronous wait follows every reflected stage, only finitely many reflections occur on any bounded time interval. Thus no finite-time definition involves an infinite product of reflections. The proof reduces the marginal law on each bounded interval to the standard Cameron–Martin reflection coupling used by the source paper, concatenated at stopping times.

The convergence estimate deliberately avoids moments of A_n , whose expectation is infinite. It uses only the tail bound $\mathbb{P}[A_n \geq M] \leq M^{-1}$ and the summability of $\|h_n\|_{\mathcal{W}}$.

6 Novelty check

Before promotion, I searched the run indexes

`registry_index.tsv`, `solutions/index.tsv`, `attempts/index.tsv`, `proof_gaps/index.tsv`

for the arXiv id, title, and core phrases including `coupling at time infinity`, `abstract Wiener space`, and `Brownian coupling`. No existing packet or attempt answered this target.

I also searched the local parsed source corpus for the exact coupling phrases and close abstract-Wiener keywords. The only relevant occurrences were the source paper and target-pool metadata. A bounded web search using the queries

```
"coupling at time infinity" "abstract Wiener space",  
  "Brownian coupling at time infinity" "Banach",  
"Coupling of Brownian motions in Banach spaces" "open question",  
"Cameron-Martin" "coupling at time infinity" Brownian
```

and close variants found only the source arXiv page as a relevant hit. This is therefore recorded as a candidate original proof, not as an established literature answer. Novelty confidence remains bounded-search confidence, pending specialist review.

7 Human review recommendation

Review as a high-priority candidate full solution to Question 18. The proof is short and appears to bypass the basis machinery in the source paper by turning the paper's ε -approximate coupling remark into an iterated construction. A reviewer should focus first on whether predictable concatenations of Cameron–Martin reflections in changing deterministic directions preserve the second process as a Banach-valued Brownian motion. If that step is accepted, the remaining approximation and Borel–Cantelli argument is elementary.

References

- [1] E. Candellero and W. S. Kendall, *Coupling of Brownian motions in Banach spaces*, arXiv:1705.08300, 2017. <https://arxiv.org/abs/1705.08300>.